

# Pneumatic End Effector Calculation Package

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## Pin Selection & Spring Selection

The full calculations of both pin & spring sizing are outlined in the attached excel calculation. This section will outline the base case assumptions and calculation methodology used to evaluate the clamping force, spring feasibility, and pin size used for the end effector design. The following calculations outline the scaled down design parameters. The same calculations can be completed given regular-scale parameters, also provided in the excel sheet detailing calculations.

### *Clamping Force:*

The following calculations outline the scaled down design parameters. The same calculations can be completed given regular-scale parameters, also provided in the excel sheet detailing calculations.

#### Assumptions:

- $m = 10\text{kg}$  (Maximum allowable mass based on allowable cylinder side loading)
- $g = 9.81\text{m/s}$
- $\mu_s = 0.8$  (Conservative estimate based on the friction coefficient between steel and rubber)

The clamping force ( $F_c$ ) is evaluated via a force balance:

$$F_c = \frac{mg}{\mu_s} = \frac{(10)(9.81)}{0.8} = 122.5N$$

### *Spring Sizing:*

To validate the spring size, based on the number of pins in action, the total deflection of each spring should not exceed the maximum compression displacement outlined in spring specifications.

Assumptions:

- $N = 6$  Pins in contact on each clamp (~40%) of pins in action at maximum clamping.

Selected Spring: 9657k5356 (McMaster-Carr) – 1.5” (38.1mm) Length Spring

- $k_s = 0.9632 \text{ kN/m}$ ,  $l_{comp} = 15\text{mm}$

$$\begin{aligned}
 x_{allowable} &> x_{max} \\
 l - l_{comp} &> x_{max} \\
 l - l_{comp} &> \frac{Fc}{Nk_s} \\
 0.0381\text{m} - 0.015\text{m} &> \frac{122.5\text{N}}{6(963.2\text{N/m})} \\
 0.0381\text{m} - 0.015\text{m} &> \frac{122.5\text{N}}{6(963.2\text{N/m})} \\
 0.0232\text{m} &> 0.0212\text{m}
 \end{aligned}$$

Therefore, this spring is sufficient for rated loading conditions evaluated on base case assumptions:

### ***Pin Sizing:***

Assumptions:

- Worst case loading (Clamping force exerted on one pin)
- Cantilevered loading with point load at the edge of the pin (Largest bending & deflection)

Pin Material Properties & Geometry (Black-Oxide Alloy Steel):

- $E = 200 \text{ GPa}$
- $\sigma_y = 415 \text{ MPa}$
- $l = 1.5'' (0.0381\text{m})$
- $d = 1/4'' (0.00635\text{m})$

Determining the maximum deflection & factor of safety of the pin, we have:

$$y = -\frac{Fl^3}{3EI} = -\frac{mgl^3}{3E\frac{\pi}{4}r^4} = -\frac{64mgl^3}{3E\pi d^4} = -\frac{64(10)(9.81)(0.0381)^3}{3(200 \cdot 10^9)\pi(0.00635)^4} = -0.113\text{mm}$$

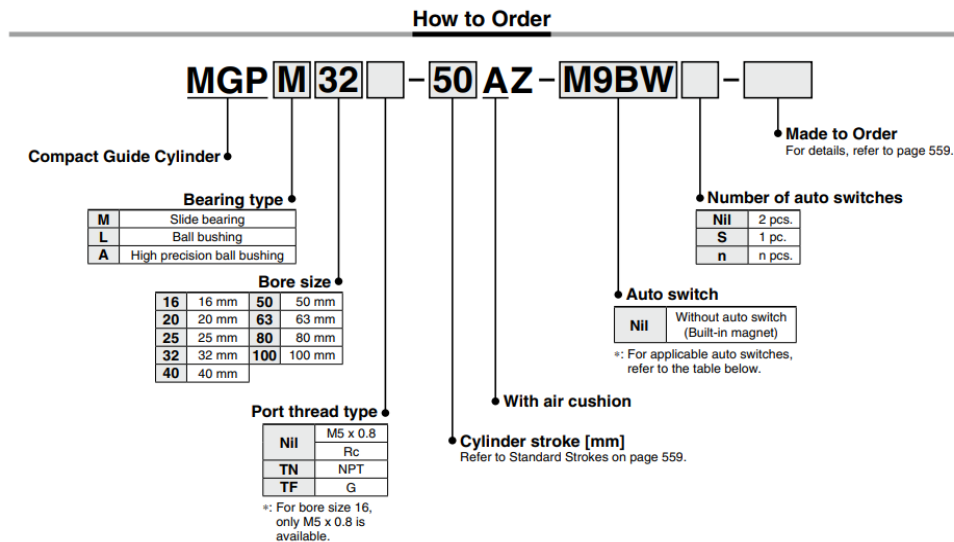
$$N = \frac{(415 \cdot 10^6)\pi(0.00635)^3}{32(10)(9.81)(0.0381)} = 2.7$$

Therefore, the maximum displacement and factor of safety of the pins under worst-case loading are within acceptable ranges.

# Pneumatic Cylinder Selection & Fluid Power Calculations

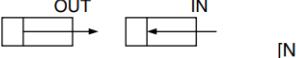
## Cylinder Type: MGP Series – Compact Guide Cylinder

- Guided cylinders with 2 acting support rods to assist in experienced bending loads
- Must be designed to be within operating load capacities based on bore size



## Theoretical Output – Desired Clamping Force

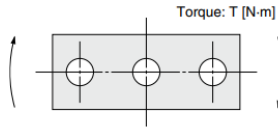
### Theoretical Output



| Bore size<br>[mm] | Rod size<br>[mm] | Operating<br>direction | Piston area<br>[mm²] | Operating pressure [MPa] |      |      |      |      |      |      |      |      |
|-------------------|------------------|------------------------|----------------------|--------------------------|------|------|------|------|------|------|------|------|
|                   |                  |                        |                      | 0.2                      | 0.3  | 0.4  | 0.5  | 0.6  | 0.7  | 0.8  | 0.9  | 1.0  |
| 12                | 6                | OUT                    | 113                  | 23                       | 34   | 45   | 57   | 68   | 79   | 90   | 102  | 113  |
|                   |                  | IN                     | 85                   | 17                       | 25   | 34   | 42   | 51   | 59   | 68   | 76   | 85   |
| 16                | 8                | OUT                    | 201                  | 40                       | 60   | 80   | 101  | 121  | 141  | 161  | 181  | 201  |
|                   |                  | IN                     | 151                  | 30                       | 45   | 60   | 75   | 90   | 106  | 121  | 136  | 151  |
| 20                | 10               | OUT                    | 314                  | 63                       | 94   | 126  | 157  | 188  | 220  | 251  | 283  | 314  |
|                   |                  | IN                     | 236                  | 47                       | 71   | 94   | 118  | 141  | 165  | 188  | 212  | 236  |
| 25                | 10               | OUT                    | 491                  | 98                       | 147  | 196  | 245  | 295  | 344  | 393  | 442  | 491  |
|                   |                  | IN                     | 412                  | 82                       | 124  | 165  | 206  | 247  | 289  | 330  | 371  | 412  |
| 32                | 14               | OUT                    | 804                  | 161                      | 241  | 322  | 402  | 483  | 563  | 643  | 724  | 804  |
|                   |                  | IN                     | 650                  | 130                      | 195  | 260  | 325  | 390  | 455  | 520  | 585  | 650  |
| 40                | 14               | OUT                    | 1257                 | 251                      | 377  | 503  | 628  | 754  | 880  | 1005 | 1131 | 1257 |
|                   |                  | IN                     | 1103                 | 221                      | 331  | 441  | 551  | 662  | 772  | 882  | 992  | 1103 |
| 50                | 18               | OUT                    | 1963                 | 393                      | 589  | 785  | 982  | 1178 | 1374 | 1571 | 1767 | 1963 |
|                   |                  | IN                     | 1709                 | 342                      | 513  | 684  | 855  | 1025 | 1196 | 1367 | 1538 | 1709 |
| 63                | 18               | OUT                    | 3117                 | 623                      | 935  | 1247 | 1559 | 1870 | 2182 | 2494 | 2806 | 3117 |
|                   |                  | IN                     | 2863                 | 573                      | 859  | 1145 | 1431 | 1718 | 2004 | 2290 | 2576 | 2863 |
| 80                | 22               | OUT                    | 5027                 | 1005                     | 1508 | 2011 | 2513 | 3016 | 3519 | 4021 | 4524 | 5027 |
|                   |                  | IN                     | 4646                 | 929                      | 1394 | 1859 | 2323 | 2788 | 3252 | 3717 | 4182 | 4646 |
| 100               | 26               | OUT                    | 7854                 | 1571                     | 2356 | 3142 | 3927 | 4712 | 5498 | 6283 | 7069 | 7854 |
|                   |                  | IN                     | 7323                 | 1465                     | 2197 | 2929 | 3662 | 4394 | 5126 | 5858 | 6591 | 7323 |

※: Theoretical output [N] = Pressure [MPa] x Piston area [mm²]

## Allowable Rotational Torque of Plate



| Bore size [mm] | Bearing type | Stroke [mm] |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | T [N-m] |  |  |  |
|----------------|--------------|-------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|---------|--|--|--|
|                |              | 10          | 20   | 25   | 30   | 40   | 50   | 75   | 100  | 125  | 150  | 175  | 200  | 250  | 300  | 350  | 400  |         |  |  |  |
| 12             | MGPM         | 0.39        | 0.32 | —    | 0.27 | 0.24 | 0.21 | 0.43 | 0.36 | 0.31 | 0.27 | 0.24 | 0.22 | 0.19 | —    | —    | —    |         |  |  |  |
|                | MGPL/A       | 0.61        | 0.45 | —    | 0.35 | 0.58 | 0.50 | 0.37 | 0.29 | 0.24 | 0.20 | 0.18 | 0.16 | 0.12 | —    | —    | —    |         |  |  |  |
| 16             | MGPM         | 0.69        | 0.58 | —    | 0.49 | 0.43 | 0.38 | 0.69 | 0.58 | 0.50 | 0.44 | 0.40 | 0.36 | 0.30 | —    | —    | —    |         |  |  |  |
|                | MGPL/A       | 0.99        | 0.74 | —    | 0.59 | 0.99 | 0.86 | 0.65 | 0.52 | 0.43 | 0.37 | 0.32 | 0.28 | 0.23 | —    | —    | —    |         |  |  |  |
| 20             | MGPM         | —           | 1.05 | —    | 0.93 | 0.83 | 0.75 | 1.88 | 1.63 | 1.44 | 1.28 | 1.16 | 1.06 | 0.90 | 0.78 | 0.69 | 0.62 |         |  |  |  |
|                | MGPL/A       | —           | 1.26 | —    | 1.03 | 2.17 | 1.94 | 1.52 | 1.25 | 1.34 | 1.17 | 1.03 | 0.93 | 0.76 | 0.65 | 0.56 | 0.49 |         |  |  |  |
| 25             | MGPM         | —           | 1.76 | —    | 1.55 | 1.38 | 1.25 | 2.96 | 2.57 | 2.26 | 2.02 | 1.83 | 1.67 | 1.42 | 1.24 | 1.09 | 0.98 |         |  |  |  |
|                | MGPL/A       | —           | 2.11 | —    | 1.75 | 3.37 | 3.02 | 2.38 | 1.97 | 2.05 | 1.78 | 1.58 | 1.41 | 1.16 | 0.98 | 0.85 | 0.74 |         |  |  |  |
| 32             | MGPM         | —           | —    | 6.35 | —    | —    | 5.13 | 5.69 | 4.97 | 4.42 | 3.98 | 3.61 | 3.31 | 2.84 | 2.48 | 2.20 | 1.98 |         |  |  |  |
|                | MGPL/A       | —           | —    | 5.95 | —    | —    | 4.89 | 5.11 | 4.51 | 6.34 | 5.79 | 5.33 | 4.93 | 4.29 | 3.78 | 3.38 | 3.04 |         |  |  |  |
| 40             | MGPM         | —           | —    | 7.00 | —    | —    | 5.66 | 6.27 | 5.48 | 4.87 | 4.38 | 3.98 | 3.65 | 3.13 | 2.74 | 2.43 | 2.19 |         |  |  |  |
|                | MGPL/A       | —           | —    | 6.55 | —    | —    | 5.39 | 5.62 | 4.96 | 6.98 | 6.38 | 5.87 | 5.43 | 4.72 | 4.16 | 3.71 | 3.35 |         |  |  |  |
| 50             | MGPM         | —           | —    | 13.0 | —    | —    | 10.8 | 12.0 | 10.6 | 9.50 | 8.60 | 7.86 | 7.24 | 6.24 | 5.49 | 4.90 | 4.43 |         |  |  |  |
|                | MGPL/A       | —           | —    | 9.17 | —    | —    | 7.62 | 9.83 | 8.74 | 11.6 | 10.7 | 9.83 | 9.12 | 7.95 | 7.02 | 6.26 | 5.63 |         |  |  |  |
| 63             | MGPM         | —           | —    | 14.7 | —    | —    | 12.1 | 13.5 | 11.9 | 10.7 | 9.69 | 8.86 | 8.16 | 7.04 | 6.19 | 5.52 | 4.99 |         |  |  |  |
|                | MGPL/A       | —           | —    | 10.2 | —    | —    | 8.48 | 11.0 | 9.74 | 13.0 | 11.9 | 11.0 | 10.2 | 8.84 | 7.80 | 6.94 | 6.24 |         |  |  |  |
| 80             | MGPM         | —           | —    | 21.9 | —    | —    | 18.6 | 22.9 | 20.5 | 18.6 | 17.0 | 15.6 | 14.5 | 12.6 | 11.2 | 10.0 | 9.11 |         |  |  |  |
|                | MGPL/A       | —           | —    | 15.1 | —    | —    | 23.3 | 22.7 | 20.6 | 18.9 | 17.3 | 16.0 | 14.8 | 12.9 | 11.3 | 10.0 | 8.94 |         |  |  |  |
| 100            | MGPM         | —           | —    | 38.8 | —    | —    | 33.5 | 37.5 | 33.8 | 30.9 | 28.4 | 26.2 | 24.4 | 21.4 | 19.1 | 17.2 | 15.7 |         |  |  |  |
|                | MGPL/A       | —           | —    | 27.1 | —    | —    | 30.6 | 37.9 | 34.6 | 31.8 | 29.3 | 27.2 | 25.3 | 22.1 | 19.5 | 17.3 | 15.5 |         |  |  |  |

- In analysis, loading is assumed to be perfectly aligned with the center of actuation, meaning minimal to no plate torque is generated from operation
- In previous designs, eccentricity (>100mm) would have caused resultingly high plate torque making the design unsuitable for using even guided cylinders

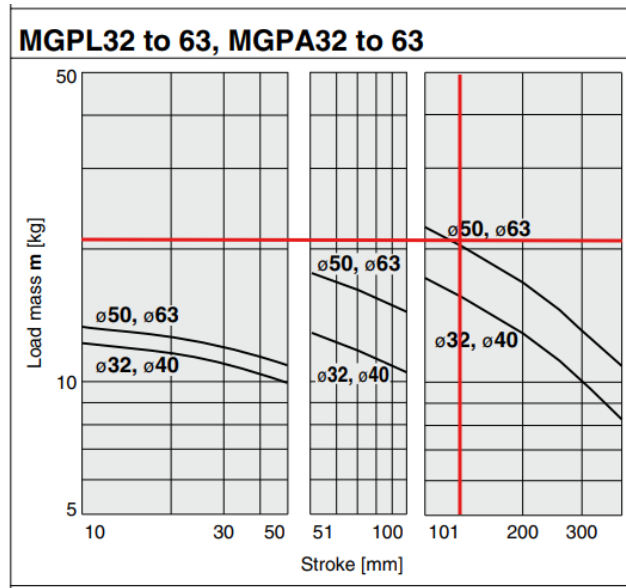
## Selection Conditions

| Mounting orientation  | Vertical    |             | Horizontal  |            |
|-----------------------|-------------|-------------|-------------|------------|
|                       |             |             |             |            |
| Maximum speed [mm/s]  | 200 or less | 400         | 200 or less | 400        |
| Graph (Slide bearing) | (1), (2)    | (3), (4)    | (13), (14)  | (15), (16) |
| Graph (Ball bushing)  | (5) to (8)  | (9) to (12) | (17), (18)  | (19), (20) |

- To determine if loading is appropriate given limitations to cylinder, horizontal loading under 200mm/s is required. Slow actuation is desirable to withstand inertial forces acting on the mechanism.

## Regular Scale: MPGL-50-130Z

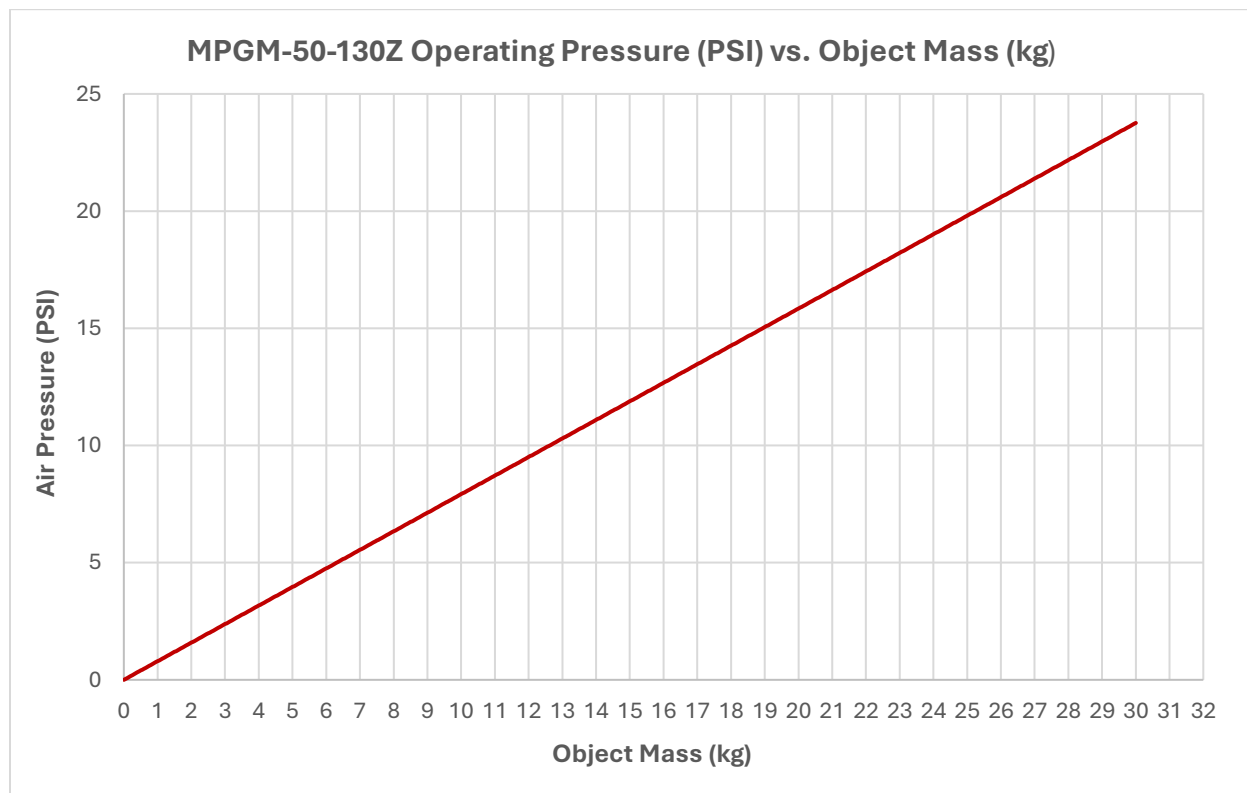
- Stroke length = 130mm, Bore Diameter = 50mm, Piston Diameter = 18mm
- Maximum Load Mass = 21kg
- For appropriate design, maximum allowable mass per arm = 20kg (-1kg)
- Max Operating Part Mass =  $2 * (20 - 5) = 30\text{kg}$
- **Thus, the maximum desired part mass can be operated using MPGM-50-130Z.**



To determine the operating pressure for any mass,  $m$ , the following relationship is formed:

$$P = \frac{F_c}{A_{bore}} = \frac{145.038 \cdot \left(\frac{9.8m}{\mu}\right)}{10^6 \cdot \left(\frac{\pi}{4}\right) \cdot ((d_{bore})^2 - (d_{rod})^2)} = \frac{145.038 \cdot \left(\frac{9.8m}{0.8}\right)}{10^6 \cdot \left(\frac{\pi}{4}\right) \cdot ((0.05)^2 - (0.018)^2)} = 0.8m$$

$$P(\text{psi}) = 0.8 \cdot m \text{ (kg)}$$

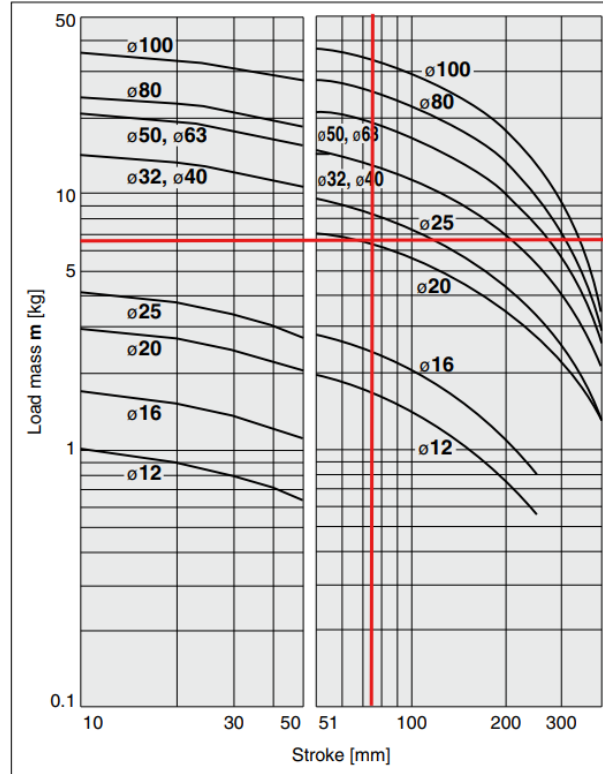


### Scaled Down: MPGM-20-75Z

- Stroke length = 75mm, Bore Diameter = 20mm, Rod Diameter = 10mm
- Maximum Load Mass = 6.75kg
- For appropriate design, maximum allowable mass per arm = 5.75kg (-1kg)
- Max Operating Part Mass =  $2 \cdot (5.75\text{kg} - \text{Mass of Arm}) = 2 \cdot (5.75 - 0.75) = 10\text{kg}$
- **Thus, the maximum desired part mass can be operated using MPGM-20-75Z.**

### MGPM12 to 100

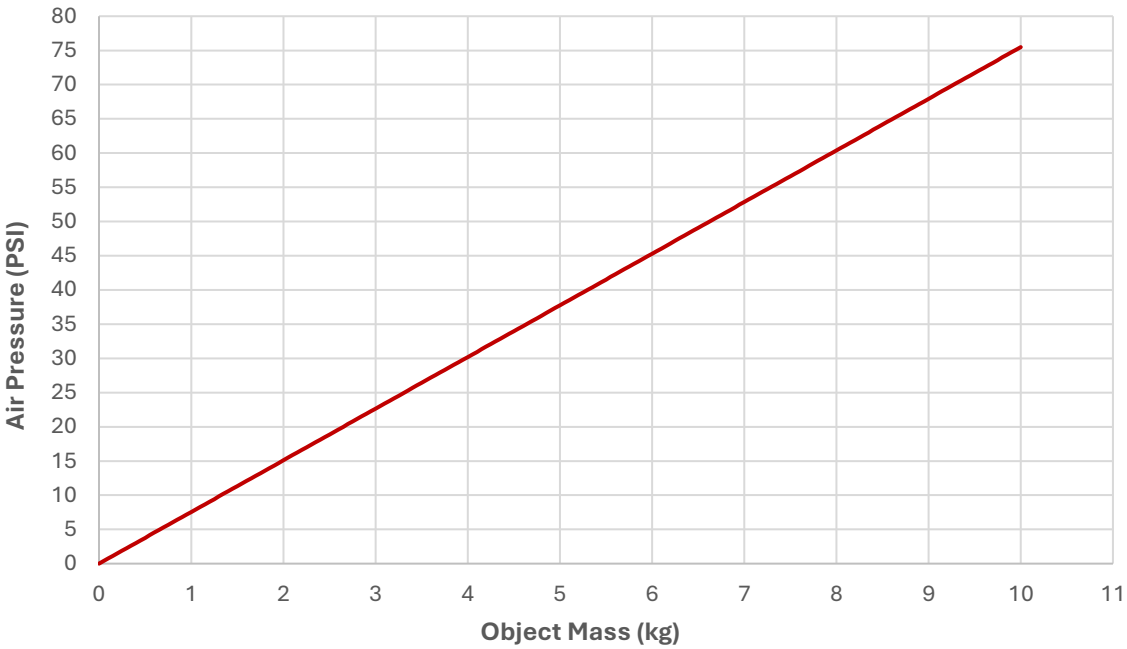
(13) L = 50 mm, V = 200 mm/s or less



$$P = \frac{F_c}{A_{bore}} = \frac{145.038 \cdot \left(\frac{9.8m}{\mu}\right)}{10^6 \cdot \left(\frac{\pi}{4}\right) \cdot ((d_{bore})^2 - (d_{rod})^2)} = \frac{145.038 \cdot \left(\frac{9.8m}{0.8}\right)}{10^6 \cdot \left(\frac{\pi}{4}\right) \cdot ((0.02)^2 - (0.01)^2)} = 7.54m$$

$$P(\text{psi}) = 7.54 \cdot m (\text{kg})$$

MPGM-20-75Z Operating Pressure (PSI) vs. Object Mass (kg)



## Cylinder Clamp Connection Weldment Structural Analysis Results

- Material: Aluminum 6061 Alloy ( $\sigma_Y = 280\text{MPa}$ )
- Loading: Worst-case loading along centre of plate, 1.25x maximum load, bottom of frame
- Meshing: Generally lowest allowable element size before size limit is achieved

### Preliminary Design: Regular Scale – Both Arms

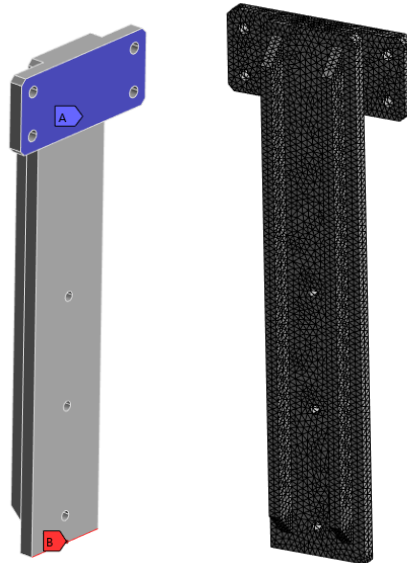
| Stress Analysis Results                                       |               |
|---------------------------------------------------------------|---------------|
| Maximum Stress (MPa)                                          | 32 (SF = 8.5) |
| <b>Max Deflection (mm)</b>                                    | <b>1.2</b>    |
| Max Cycles (Zero Based, Goodman), Worst Case Loading (1.25Fc) | N = $10^8$    |

### Boundary/Loading Conditions, Meshing

- Assumed loading through center of plate, line pressure placed at bottom of plate for worst-case bending stress & max deflection; loading at rated clamping force (No force FOS applied)
- Fixed supports from back of cylinder plate
- Fillet welds not added, stress concentrations will result in higher stresses (worst-case)

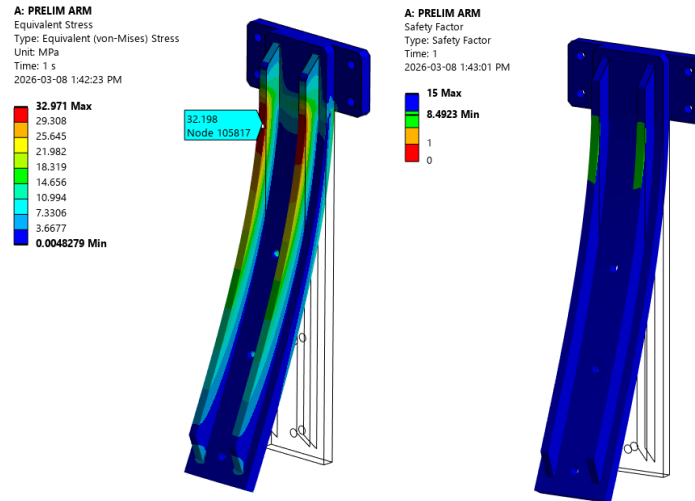
A: PRELIM ARM  
Static Structural  
Time: 1. s  
2026-03-08 1:49:27 PM

A Fixed Support  
B Force: 420. N

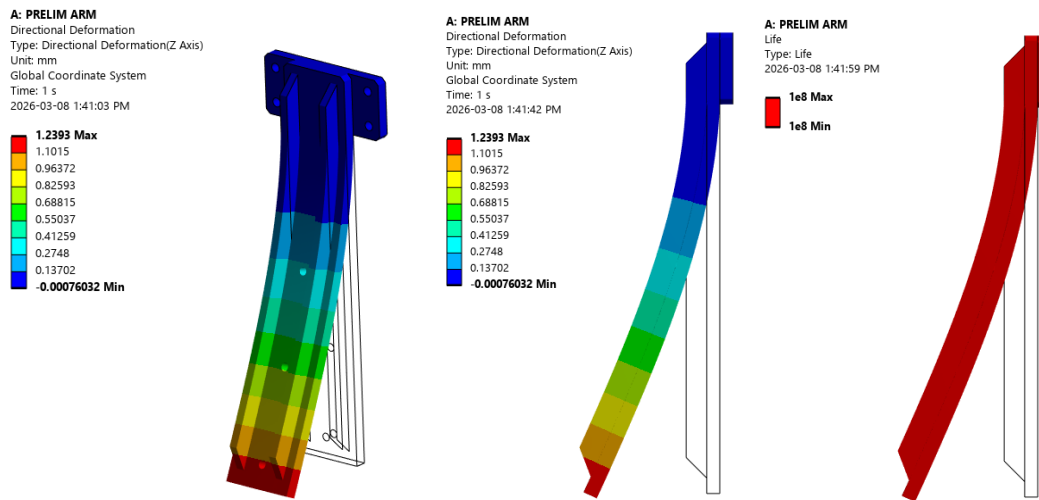


| Details of "Body Sizing" - Sizing |                    |
|-----------------------------------|--------------------|
| Scope                             |                    |
| Scoping Method                    | Geometry Selection |
| Geometry                          | 1 Body             |
| Definition                        |                    |
| Suppressed                        | No                 |
| Type                              | Element Size       |
| Element Size                      | 5.0 mm             |
| Advanced                          |                    |
| Defeature Size                    | Default            |
| Behavior                          | Soft               |

### Equivalent (Von-Mises) Stress Results



## Deflection & Cyclic Loading Results



## V1: Regular Scale – Long Arm

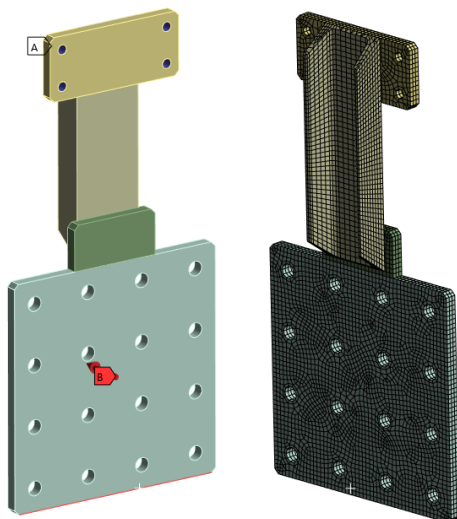
| Stress Analysis Results                                       |                   |
|---------------------------------------------------------------|-------------------|
| Maximum Stress (MPa)                                          | 66 (SF = 4.2)     |
| <b>Max Deflection (mm)</b>                                    | <b>2.01</b>       |
| Max Cycles (Zero Based, Goodman), Worst Case Loading (1.25Fc) | N=10 <sup>8</sup> |

## Boundary/Loading Conditions, Meshing

- Assumed loading through center of plate, line pressure placed at bottom of plate for worst-case bending stress & max deflection; loading at ~1.25x maximum clamping force (~320N)
- Fixed supports from cylinder plate hole locations
- Fillet welds not added, stress concentrations will result in higher stresses (worst-case)

**B: V1 ARM-FULL SCALE**  
 Static Structural  
 Time: 1 s  
 2026-03-08 1:26:51 PM

**A** Fixed Support  
**B** Remote Force: -400. N

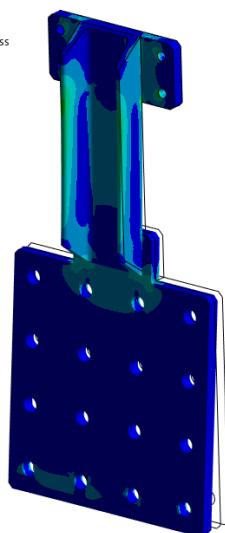


| Details of "Body Sizing" - Sizing       |                    |
|-----------------------------------------|--------------------|
| <b>Scope</b>                            |                    |
| Scoping Method                          | Geometry Selection |
| Geometry                                | 4 Bodies           |
| <b>Definition</b>                       |                    |
| Suppressed                              | No                 |
| Type                                    | Element Size       |
| <input type="checkbox"/> Element Size   | 5.0 mm             |
| <b>Advanced</b>                         |                    |
| <input type="checkbox"/> Defeature Size | Default            |
| Behavior                                | Soft               |

## Equivalent (Von-Mises) Stress Results

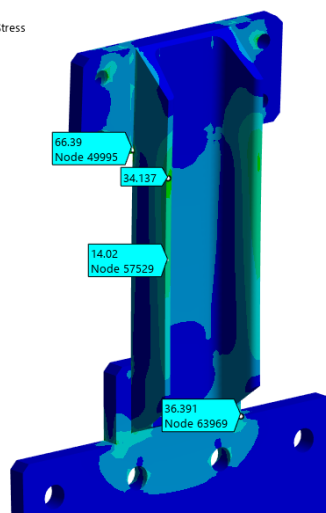
**B: V1 ARM-FULL SCALE**  
 Equivalent Stress  
 Type: Equivalent (von-Mises) Stress  
 Unit: MPa  
 Time: 1 s  
 2026-03-08 1:27:21 PM

67.158  
**66.39 Max**  
 58.766  
 50.374  
 41.982  
 33.59  
 25.199  
 16.807  
 8.4149  
**0.023101 Min**

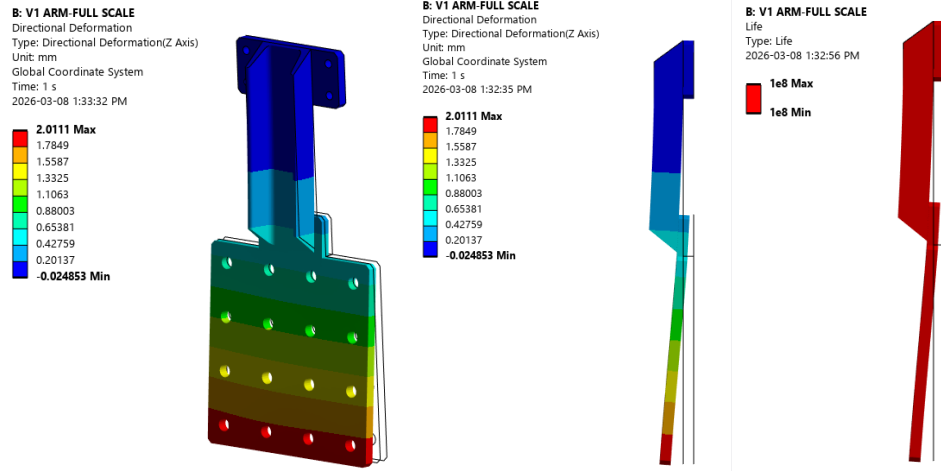


**B: V1 ARM-FULL SCALE**  
 Equivalent Stress  
 Type: Equivalent (von-Mises) Stress  
 Unit: MPa  
 Time: 1 s  
 2026-03-08 1:29:47 PM

67.158  
**66.39 Max**  
 58.766  
 50.374  
 41.982  
 33.59  
 25.199  
 16.807  
 8.4149  
**0.023101 Min**



## Deflection & Cyclic Loading Results



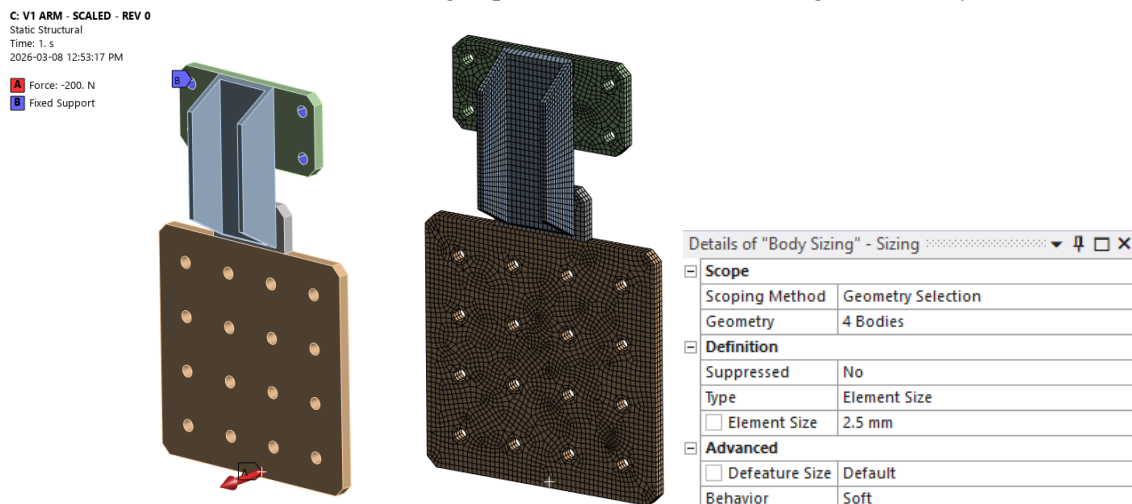
## V1: Scaled Down

### Dynamic Weldment

| Stress Analysis Results                                      |                 |
|--------------------------------------------------------------|-----------------|
| Maximum Stress (MPa)                                         | 184 (SF = 1.52) |
| <b>Max Deflection (mm)</b>                                   | <b>2.05</b>     |
| Max Cycles (Zero Based, Goodman), Worst Case Loading (1.5Fc) | N=1,167,200     |

### Boundary/Loading Conditions, Meshing

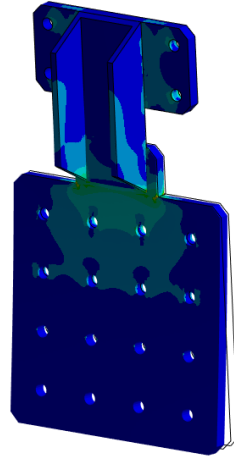
- Assumed loading through center of plate, line pressure placed at bottom of plate for worst-case bending stress & max deflection; loading at ~1.5x maximum clamping force (~125N)
- Fixed supports from cylinder plate hole locations
- Does not contain a fillet, increasing expected stresses and reducing available cycles



### Equivalent (Von-Mises) Stress Results

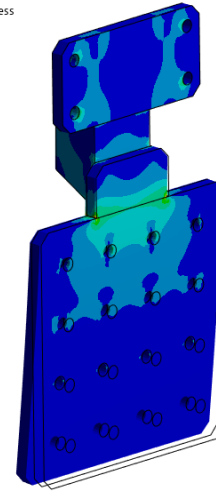
C: V1 ARM - SCALED - REV 0  
Equivalent Stress  
Type: Equivalent (von-Mises) Stress  
Unit: MPa  
Time: 1 s  
2026-03-08 12:55:00 PM

183.96 Max  
163.53  
143.1  
122.66  
102.23  
81.801  
61.37  
40.939  
20.508  
0.076344 Min



C: V1 ARM - SCALED - REV 0  
Equivalent Stress  
Type: Equivalent (von-Mises) Stress  
Unit: MPa  
Time: 1 s  
2026-03-08 12:57:44 PM

183.96 Max  
163.53  
143.1  
122.66  
102.23  
81.801  
61.37  
40.939  
20.508  
0.076344 Min



## Deflection & Cyclic Loading Results

C: V1 ARM - SCALED - REV 0  
Directional Deformation  
Type: Directional Deformation(X Axis)  
Unit: mm  
Global Coordinate System  
Time: 1 s  
2026-03-08 1:00:46 PM

0.034395 Max  
-0.1956  
-0.42559  
-0.65558  
-0.88557  
-1.1156  
-1.3456  
-1.5755  
-1.8055  
-2.0355 Min



C: V1 ARM - SCALED - REV 0  
Life  
Type: Life  
2026-04-03 2:20:44 PM

1e8 Max  
6.0987e7  
3.7194e7  
2.2684e7  
1.3834e7  
8.437e6  
5.1455e6  
3.1381e6  
1.9138e6  
1.1672e6 Min



## Static Weldment

- Fillet weld added on clamp plate connection piece to lower stress concentration
- Angle thickness increased from 0.25 to 0.5in to reduce bending stress on component

| Stress Analysis Results                                      |                 |
|--------------------------------------------------------------|-----------------|
| Maximum Stress (MPa)                                         | 137 (SF = 2.05) |
| Max Deflection (mm)                                          | 2.85            |
| Max Cycles (Zero Based, Goodman), Worst Case Loading (1.5Fc) | N = 65,686,000  |

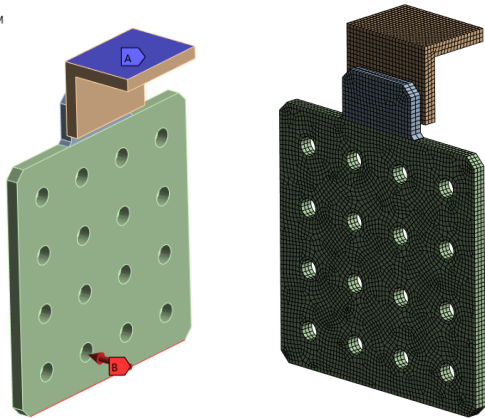
## Boundary/Loading Conditions, Meshing

- Assumed loading through center of plate, line pressure placed at bottom of plate for worst-case bending stress & max deflection; loading at ~1.5x maximum clamping force (~125N)

D: Copy of V1 ARM - SCALED - REV 1 - FILLET WELD ADDED

Static Structural  
Time: 1 s  
2026-03-08 12:35:46 PM

**A** Fixed Support  
**B** Force: 200. N



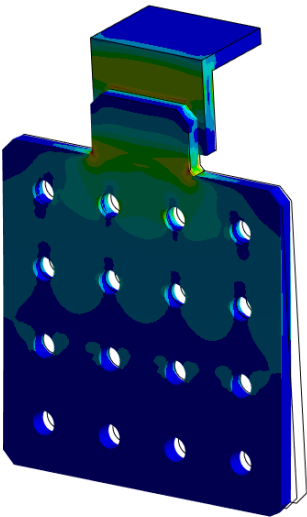
| Details of "Face Sizing" - Sizing       |                          |
|-----------------------------------------|--------------------------|
| [-] Scope                               |                          |
| Scoping Method                          | Geometry Selection       |
| Geometry                                | 44 Faces                 |
| [-] Definition                          |                          |
| Suppressed                              | No                       |
| Type                                    | Element Size             |
| <input type="checkbox"/> Element Size   | 2.0 mm                   |
| [-] Advanced                            |                          |
| <input type="checkbox"/> Defeature Size | Default (5.4491e-002 mm) |
| Influence Volume                        | No                       |
| Behavior                                | Soft                     |
| <input type="checkbox"/> Growth Rate    | Default (1.85)           |
| Capture Curvature                       | No                       |
| Capture Proximity                       | No                       |

Equivalent (Von-Mises) Stress Results

D: Copy of V1 ARM - SCALED - REV 1 - FILLET WELD ADDED

Equivalent Stress  
Type: Equivalent (von-Mises) Stress  
Unit: MPa  
Time: 1 s  
2026-03-07 4:04:43 PM

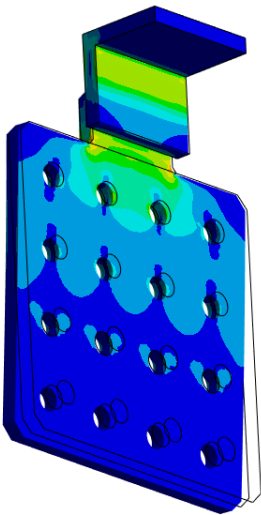
**136.58 Max**  
121.4  
106.23  
91.054  
75.879  
60.704  
45.529  
30.354  
15.179  
**0.0036622 Min**



D: Copy of V1 ARM - SCALED - REV 1 - FILLET WELD ADDED

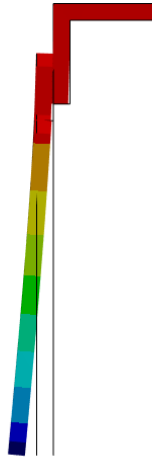
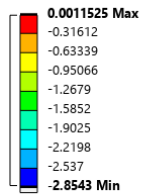
Equivalent Stress  
Type: Equivalent (von-Mises) Stress  
Unit: MPa  
Time: 1 s  
2026-03-07 4:04:43 PM

**136.58 Max**  
121.4  
106.23  
91.054  
75.879  
60.704  
45.529  
30.354  
15.179  
**0.0036622 Min**

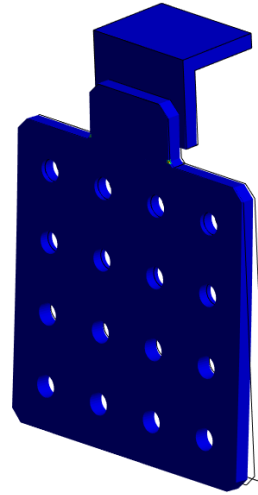
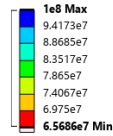


Deflection & Cyclic Loading Results

D: Copy of V1 ARM - SCALED - REV 1 - FILLET WELD ADDED  
 Directional Deformation  
 Type: Directional Deformation(X Axis)  
 Unit: mm  
 Global Coordinate System  
 Time: 1 s  
 2026-03-07 4:15:54 PM



D: Copy of V1 ARM - SCALED - REV 1 - FILLET WELD ADDED  
 Life  
 Type: Life  
 2026-04-03 2:22:01 PM



## Structural Analysis Results (ANSYS) Summary

|                                                              | Preliminary Design | Refined Design Long Arm | Refined Design (Scaled) Static | Refined Design (Scaled) Dynamic |
|--------------------------------------------------------------|--------------------|-------------------------|--------------------------------|---------------------------------|
| <b>Loading Condition</b>                                     | 1.25Fc             | 1.25Fc                  | 1.5Fc                          | 1.5Fc                           |
| Maximum Stress (MPa)                                         | 32                 | 66                      | 137                            | 184                             |
| Safety Factor (SF)                                           | 8.5                | 4.2                     | 2.05                           | 1.52                            |
| Max Deflection (mm)                                          | 1.2                | 2.01                    | 2.85                           | 2.05                            |
| Max Cycles (Zero Based, Goodman), Worst Case Loading (1.5Fc) | N >10e8            | N >10e8                 | N = 65,686,000                 | N=1,167,200                     |